

JAGRITI TRIPATHI

Software Engineer

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Location: Delhi, India

PROFESSIONAL SUMMARY

Software Engineer with 1.5 years shipping high-scale distributed backend systems for enterprise trading platforms processing 100k+ trades daily. Strong foundation in Java, Spring Boot, React, SQL, distributed systems, REST APIs, and DevOps practices. Proven record in performance optimization, production support, and CI/CD, with hands-on GenAI/RAG projects and a growing focus on AI-augmented engineering.

WORK EXPERIENCE

Software Engineer, NOMURA (Jul 2024 – Jan 2026 | 1.5 years)

Project: Trade Distribution Platform (Middle Office)

- Built high-performance search dashboards, optimizing existing backend APIs in **Java/Spring Boot**, reduced response latency from ~2 minutes to ~20 seconds (by ~83%), achieved through query profiling, index restructuring.
- Developed broker onboarding workflows for a microservice that processes incoming broker files consisting up to 100k+ trades into standardized **PDM/XML** formats and automated file-summary deliveries using **Bash & Autosys**.
- Improved data consistency and validation across front-office trade processing pipelines, reducing downstream recon issues.
- Replaced SSH-based log triage with a centralised Grafana + Prometheus + OpenTelemetry observability stack, reducing incident investigation from manual host lookups to instant PromQL queries across logs, metrics, and service health.
- Provided **L3 production support**, performing end-to-end triage across inbound files, relational DBs, and TIBCO EMS queues, including DLQ analysis and message replay.
- Spearheaded complete rewrite of a legacy web application from React15 → React 18, reducing load time by ~80% (2.5 min → <30s) while enhancing UX, responsiveness, and cross-team usability; successfully adopted in production.
- Configured and integrated **NGINX** as a reverse proxy for web service, enabling frontend and backend services to be served under organization-specific context paths for intranet deployment while tuning gateway timeout policies to prevent long-running DB requests.
- Migrated containerized microservices from **Docker Swarm (RHEL7)** to **Podman (RHEL8.10)** using **Quadlets and rootless systemd units** for stable enterprise deployments.
- Authored and maintained **JUnit/Mockito tests**, improving overall code reliability by achieving ~85% code coverage on SonarQube, strengthening **GitLab CI/CD** pipelines.

Technology Intern (Developer), NOMURA (Jan 2024 – Jun 2024 | 6 months)

- Designed and developed a modern web frontend end-to-end using **React 18**, upgrading legacy UI components.
- Integrated third-party APIs using **Spring Boot** for secure data retrieval for the Trade Service platform.
- Worked on a mini **web-scraping** project to extract and process structured data.

EDUCATION

B.Tech in Electronics and Communication Engineering - CGPA: 7.47

Indira Gandhi Delhi Technical University for Women, Delhi

2020-2024

XII (PCM, CBSE) - Percentage: 90.8%

School Of Excellence, Rohini

2018-2020

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, Bash, XML/XSLT, SQL (PostgreSQL, Oracle SQL)

Backend: Spring Boot, Microservices, REST APIs, Apache Camel, Event-Driven Architecture, TIBCO EMS

Frontend: React, Material UI, Bootstrap, Tailwind CSS, Gradio

GenAI: Pydantic, Prompt Engineering, LLMs, RAG, Vector Embeddings, VectorDBs(Chroma/FAISS), Hugging Face, PyTorch

DevOps & Infra: Docker, Podman, GitLab CI/CD, GitHub, Linux, AutoSys, Postman, NGINX

Observability & Testing: Grafana, Prometheus, OpenTelemetry, JUnit, Mockito

Concepts: Distributed Systems, Low-Latency Design, SOLID, Design Patterns, DSA, Agile

PROJECTS

Temporal Video RAG Engine [Python | Vector DB | HuggingFace | Prompt Engineering | Gemini | Gradio] | [GitHub](#)

Built a multi-turn conversational RAG engine that indexes video transcripts into metadata-tracked vector embeddings via ChromaDB, applying strict prompt guardrails in Gemini 3.5 to deliver sub-second grounded answers with exact video timestamp links.

GetMyTix [Java | Spring Boot | ReactJS | PostgreSQL] | [GitHub](#)

Developed a high-demand ticket booking simulation designed to handle intense concurrent demand, leveraging optimistic locking, distributed conflict control, and real-time seat state orchestration to prevent overselling for booking integrity.

SourceSea [React | Typescript | NodeJS | Express | PostgreSQL] | [GitHub](#)

Engineered a full-stack web platform that transforms static bookmarks into actionable reminders, leveraging Supabase Google authentication, CRUD-based resource management, and automated notifications to ensure important webpages are never forgotten.

CERTIFICATIONS

- Databricks Accredited Generative AI Fundamentals
- IBMXGFG Generative AI & Agentic AI
- Certified for 30 Days of Google Cloud (Performed 70 assigned labs.)
- 2nd Runner-up in LeanIn Hacks (Created a disability toolkit plugin for CWSN)